Session Objectives

NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine

At the conclusion of the session, the student should be able:

1. Define & describe the mechanisms underlying the reabsorption of glucose, urea & urate

2. Define & describe the mechanisms underlying the reabsorption of proteins, oligopeptides & amino acids.

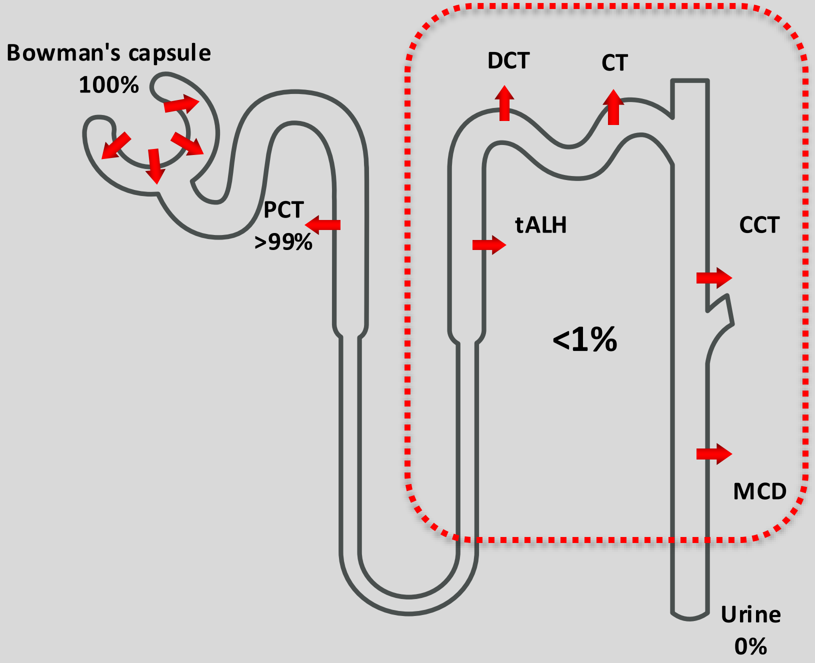
3. Define & describe the mechanisms underlying the secretion of organic acids and bases.

4. Explain the significance of the organic acid and base secretory systems.

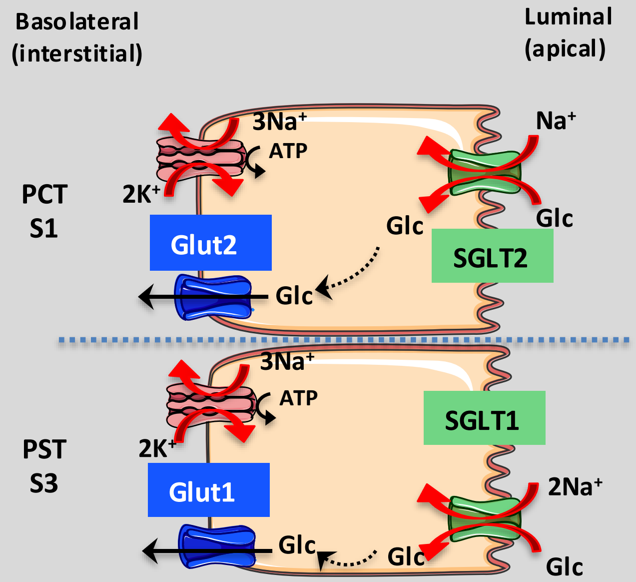
Renal Regulation of Glucose

- Plasma glucose $[Glc]_{Pl}$:
  - Remains within a narrow range (fasting Glc ≅ 4-5mM (70-100 mg/dL)).
  - Regulated by insulin & other hormones.
  - Freely filtered at the glomerulus
  - Completely reabsorbed in the PT (transcellularly).
- PCT (S1 segment):
  - Na^+-Glc cotransporter 2 (SGLT2):
    - Apical transporter
    - High-capacity, low-affinity.
    - Responsible for 90% of Glc reabsorption.
- PST (S3 segment):
  - Na^+-Glc cotransporter 1 (SGLT1):
  - Low-capacity, high-affinity,
  - Has 2:1 $Na^+ \rightarrow$ Glc stoichiometry
  - Can generate a far larger glucose gradient across the apical membrane.
- SGLT2 inhibitors are novel "glucuretics" that have been approved for the treatment of T2DM & CHF.
- Glc exits across the basolateral membrane via GLUT (Glut1 & Glut2) family of GLC channels (distinct from SGLT)
- They are Na^+-independent and move Glc via facilitated diffusion.

Glc Reabsorbtion the Nephron



Proximal Tubule



Urinary Excretion of Glucose

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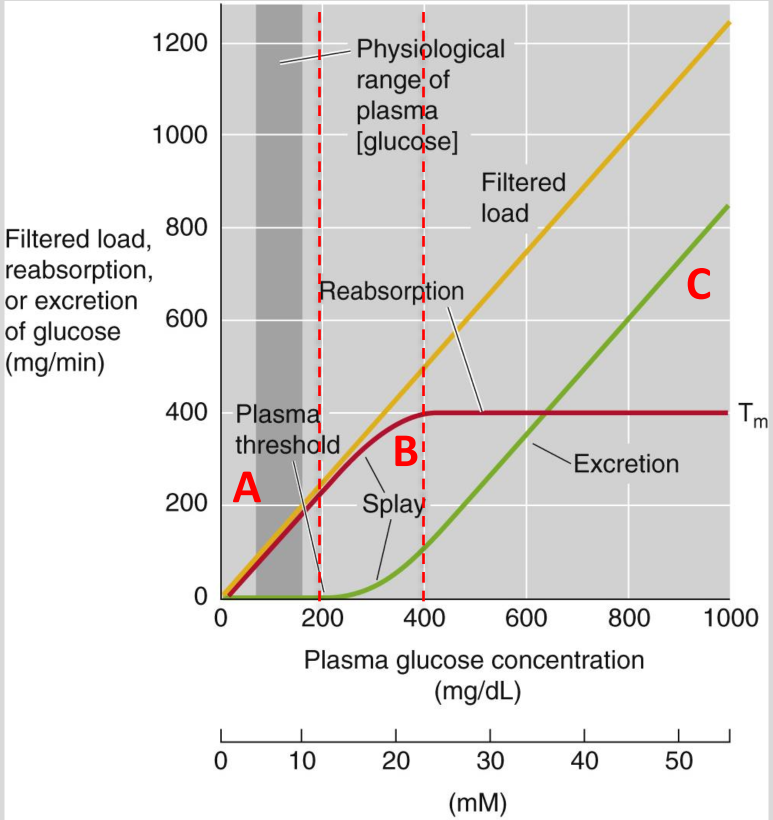
- Glucose titration curve: The relationship between $[Glc]_{Pl}$ & rate of glucose reabsorption via glucose infusion (*iv*).
- Low FL glucose (< 200 mg/dL) → no glucose in the urine → Glc_{Cl}= 0. **(A)**
- Above a threshold of ~200 mg/dL (~11 mM) Glc appears in the urine (glucosuria).
- As FL increases → ↑Glc_{Ex} (linear) → ↑Glc_{Cl} **(B)**
- At high Glc loads **(C)**, amount of Glc reabsorbed becomes small compared with the filtered load → Glc remains in the tubular lumen.
- Glc_{Exc} rises as Glc_{Pl} increases.
- Threshold (~200 mg/dL) is higher than the normal $[Glc]_{Pl}$ of ~100 mg/dL (~5.5 mM), and because the body effectively regulates $[Glc]_{Pl}$
- ∴ healthy people do not excrete any glucose in the urine, even after a meal.
- Even in people with DM, glucosuria doesn't occur until the blood sugar level exceeds threshold.
- Rate of glucose reabsorption ~ 400 mg/m → plateau → Transport maximum (T_m).
- T_{m} : SGLTs are fully saturated → Transporters can't respond to increases in Glc F.L.
- Splay: When rate of Glc reabsorption reaches the T_m gradually.
- Reflects both anatomical and kinetic differences among nephrons.
- ∴ a nephron's Glc F.L. may be mismatched to its capacity to reabsorb glucose (e.g., a nephron with a larger glomerulus has a larger load of glucose to reabsorb, different nephrons may have different distributions and densities of SGLT2 and SGLT1 along the PT.

Glc Titration Curve



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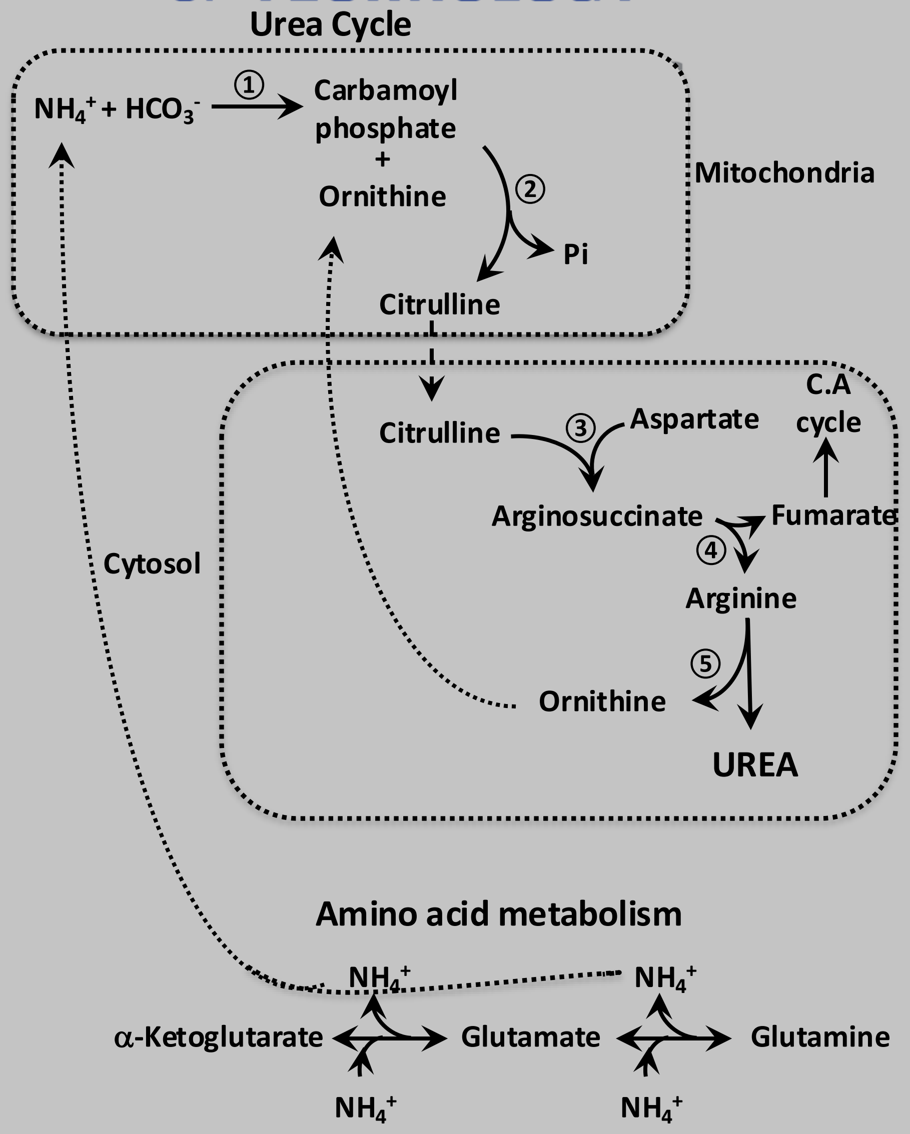
Renal Regulation of Urea

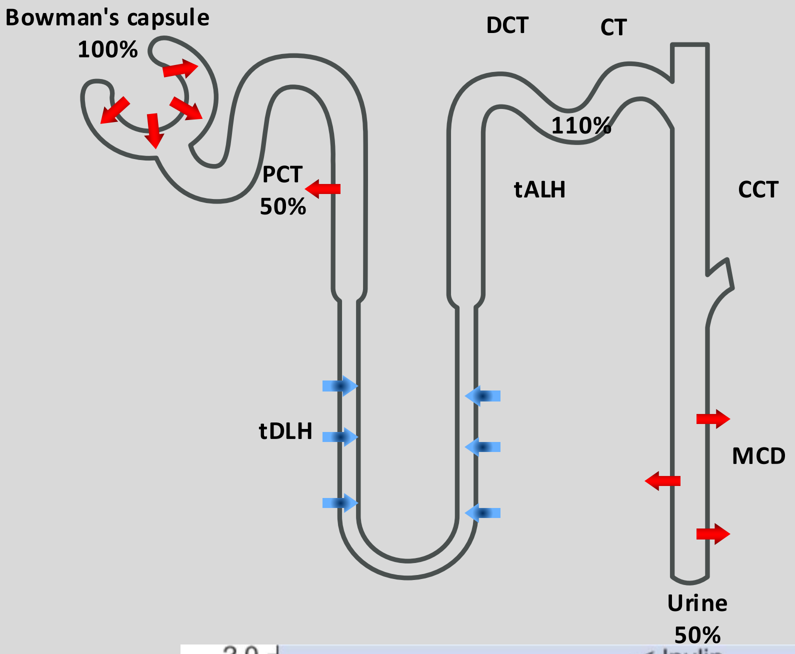
- Urea cycle: Responsible for the removal of excess N and NH_4^+ (ammonium).
  - Detoxifies ~95% of the NH_4^+ (liver).
  - Generates urea from NH_4^+, the primary nitrogenous end product of amino-acid catabolism.
- Urea (Ur) is a very special substance for the kidney.
  - MW: 60 Da, highly polar & H_2O soluble.
  - Normal plasma concentration of urea: 2.5 to 6 mM.
  - End product of protein metabolism, a waste product to be excreted, & critical for H_2O regulation and salt reabsorption.
  - Provides a critical means of ridding the body of N waste.
- Primary route of excretion: urine (some urea exit via stool & sweat).
- Freely filtered at the glomerulus, and then reabsorbed and secreted.
- Tubules reabsorb more urea than they secrete
- ∴ amount of urea excreted in the urine < amount filtered.
- Normal levels of urea are not toxic
- Large amounts are produced in a high-protein diet, representing a large osmotic load that must be excreted.
- Clinical laboratories report plasma urea levels as <u>blood urea nitrogen (BUN)</u>
- Normal BUN values: 7 to 18 mg/dL.

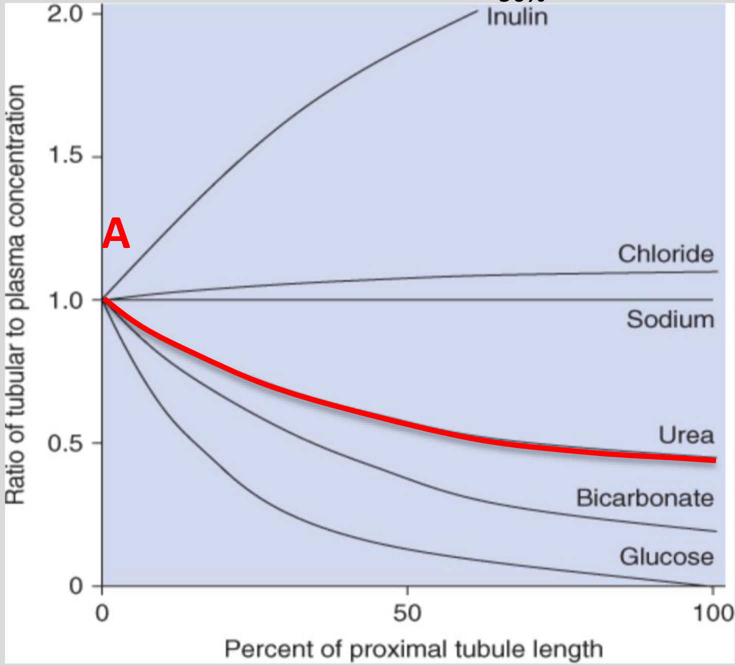


Urea Transport in the Nephron

- Urea is not bound to plasma proteins.
- Constitutes about half of the solute content of the urine.
- Does not permeate lipid bilayers.
- Urea is both reabsorbed & secreted in the nephron.
- The tubules reabsorb more urea than they secrete
- Amount of urea excreted in the urine << Quantity filtered (~50%)
    - Reabsorption: Proximal tubule & medullary collecting duct
    - Secretion: Thin descending & ascending limbs of the loop of Henle.
- In the very early proximal tubule, $[Ur]_L = [Ur]_{Pl}$ (A)
- H_2O reabsorption increases $[Ur]_L$ in the lumen, a favorable transepithelial urea gradient drives urea reabsorption.
- Urea transport depends Δ[Ur] across the tubule epithelium
- Changes in urine flow affect renal urea handling (e.g., urea excretion rises with increasing urinary flow).
- At low urine flow, tubule reabsorbs more H_2O & $\therefore$ kidneys excrete ~15% of filtered urea.
- At high urine flow $\rightarrow$ tubules reabsorb less H_2O & urea $\rightarrow$ kidneys may excrete as much as 70% of filtered urea.
- Progressive renal disease $\rightarrow$ dec GFR $\rightarrow$ low urine flow $\rightarrow$ urea retention $\rightarrow$ $\uparrow$BUN

Urea balance the Nephron



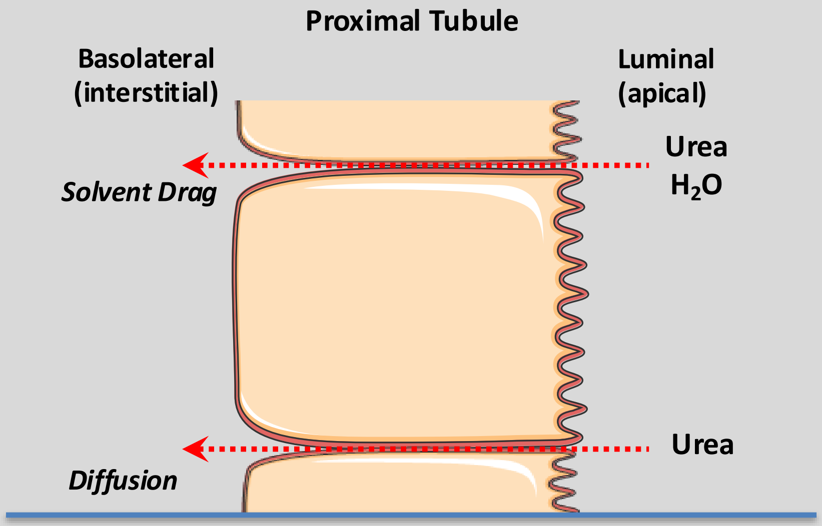


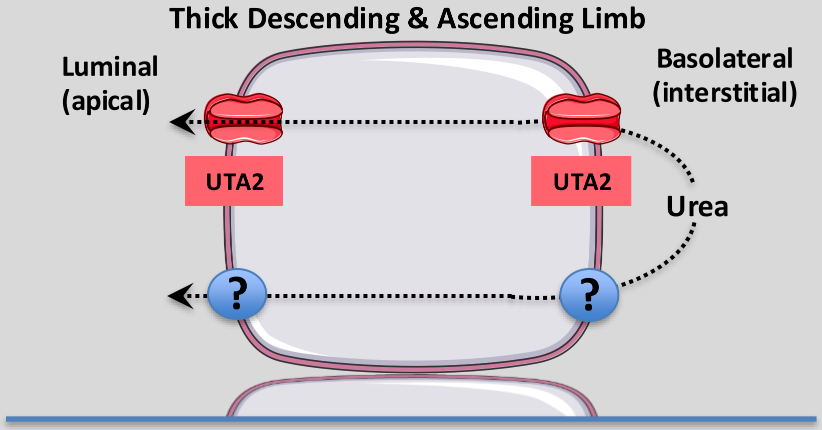
Urea Transport in the Nephron

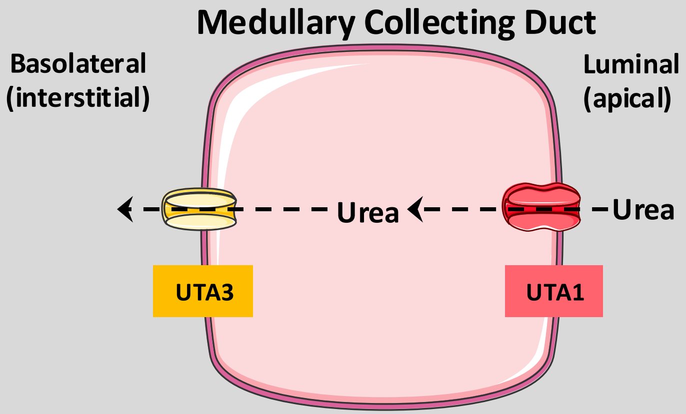
PT:
- ~ 50% of the FL of urea is reabsorbed PARACELLULARLY.
- Reabsorption occurs by solvent drag & diffusion across tight junctions.
- The greater the fluid reabsorption along the PT, the greater the reabsorption of urea.

Thin DL & AL:
- Amount equal to that reabsorbed is secreted back into the lumen of the loops of Henle (TRANSCELLULAR).
- This restores $[Ur]_L$ content to its original value.
- Secretion is via facilitated diffusion using urea transporters (UTs)
- UTs belong to the SLC14 family of solute transporters
- UTs exhibit saturation kinetics and other hallmarks of carrier-mediated transport.

MCD:
- ~ 50% of the is $[Ur]_L$ reabsorbed (TRANSCELLULAR).
- Net result: 50% of FL of urea is ultimately excreted (assuming normal urine flow & hydration status.



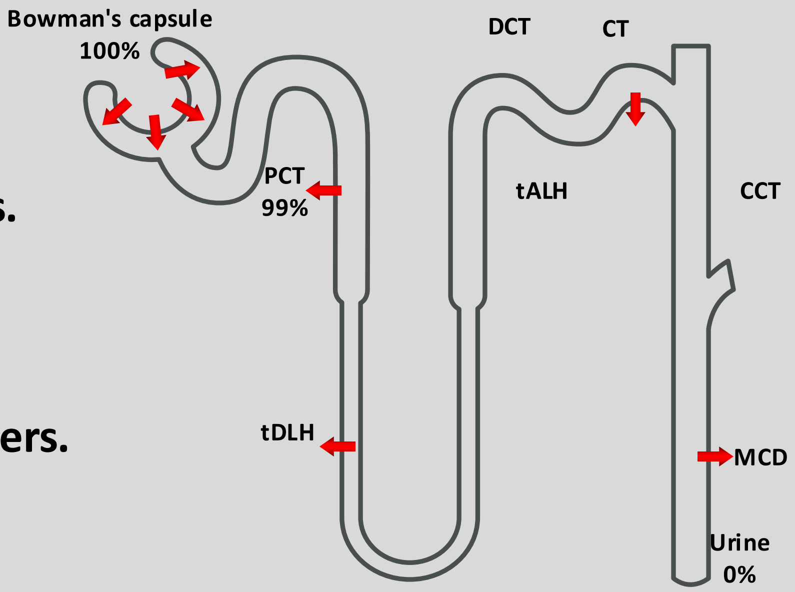


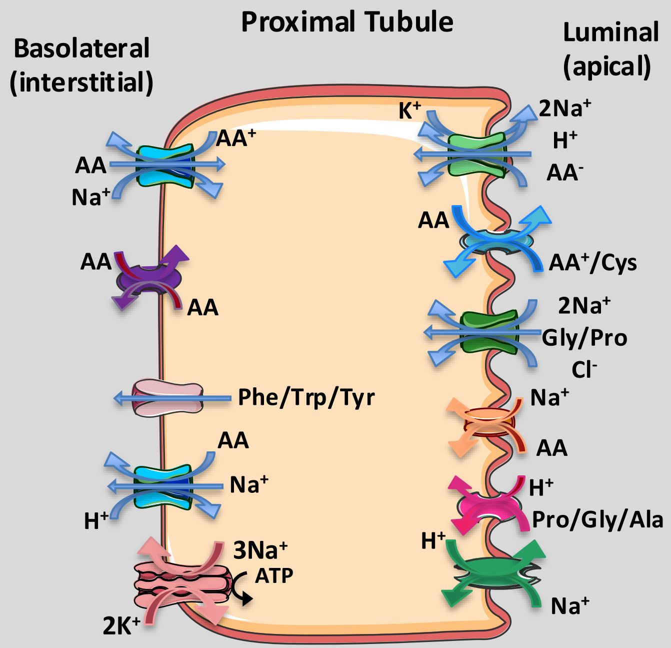


Amino Acid Transport in the Nephron

- Total concentration of amino acids (AA) in the blood is ~ 2.4 mM.
- Largely absorbed by the GI tract.
- Products of protein catabolism or *de novo* synthesis of nonessential amino acids.
- Freely filtered in the glomerulus.
- Because AA are important nutrients, they must be retrieved from the filtrate.
- PT reabsorbs >98% of AA via TRANSCELLULAR routes
- Use a wide variety of transporters of the Solute Carrier (SLC) family of transporters.
- SLCs do not have substrate specificity.
- Luminal (apical) membrane:
    - AA enters the cell via Na^+- or H^+-driven transporters
    - AA enter using AA exchangers.
- Basolateral (interstitial) membrane:
    - AA exit the cell via AA exchangers (some are Na^+-dependent)
    - Na^+-dependent uptake occurs in late PT (S3), where availability of luminal AA is low.
    - SLC38A3 mediates the transfer of AA important for cellular nutrition/metabolism across the basolateral membrane.

Amino Acid Balance in the Nephron

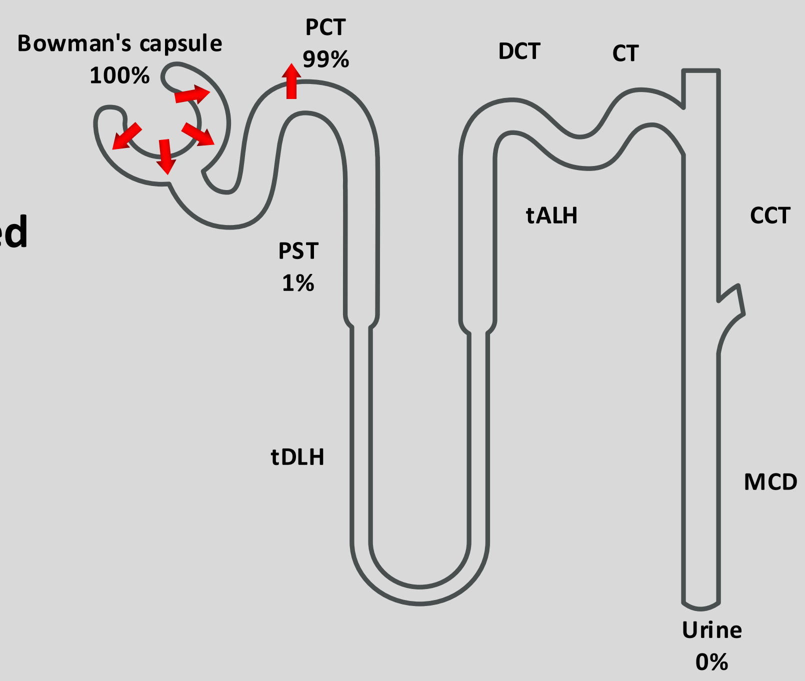




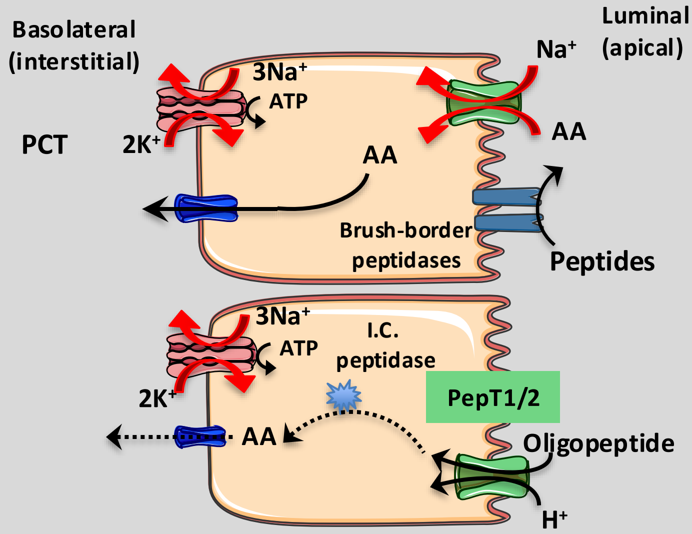
Oligopeptide Transport in the Nephron

- Oligopeptides (AA_n):
  - Short chains of AA (2-20 residues).
  - Unlike larger polypeptides or proteins, they are compact & serve specialized biological functions (e.g., signaling, immune responses, endocrine regulation).
- PT reabsorbs ~99% of filtered AA_n.
- Segments beyond the PT contribute little to peptide transport.
- Several peptidases are present at the outer surface of the <u>brush-border</u> membrane of PT cells (similar to GI tract)
- Brush-border enzymes:
  - Variety of enzymes hydrolyzing many peptides (e.g., γ-glutamyltransferase, aminopeptidases, endo- & dipeptidases)
  - Release AA into the tubule lumen
- Some AA_n (2-5 AA) are resistant to brush-border enzymes
- Enter cells via apical H^+/AA_n cotransporters PepT1/T2.
- PepT1: Low-affinity, high-capacity transporter in the PCT (S1-S2)
- PepT2: High-affinity, low-capacity transporter in the PST (S3).
- Intracellularly, AA_n are hydrolyzed by cytosolic peptidases
- Distinction between which AA_n get digested in the lumen vs I.C is not clear-cut.

Oligopeptide Reabsorption in the Nephron

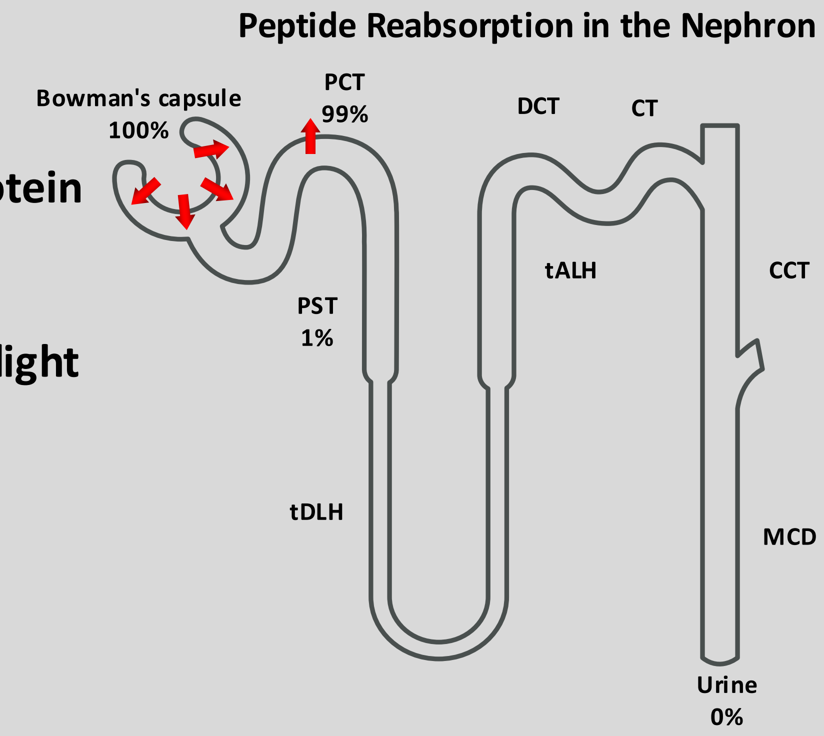


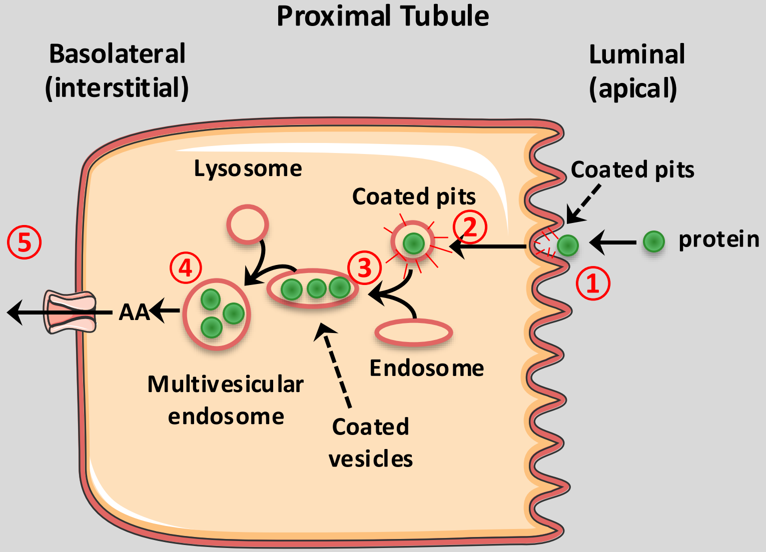
Proximal Tubule



Protein Transport in the Nephron

- The glomerular filtration barrier prevents the filtration of large amounts of protein
- This restriction is incomplete (e.g., $[Alb]_{Filtrate}$ is only 0.01-0.05% $[Alb]_{Pl}$
- Reabsorption of 96-99% of filtered albumin.
- Tubules also reabsorb relatively freely-filtered, LMW proteins (e.g., lysozyme, light chain Ig, β_2-microglobulins) & polypeptide hormones.
- ∴ tubule injury → proteinuria (even w/o glomerular injury).
- PT use receptor-mediated endocytosis to reabsorb proteins and polypeptides.
- First step 1 is binding to a receptor complex at the apical membrane.
- Followed by internalization into coated endocytic vesicles 2.
- Factors that interfere with vesicle formation/internalization (e.g., metabolic inhibitors, cytochalasin-B).
- Vesicles fuse with endosomes 3; this fusion targets the vesicle for delivery to lysosomes.
- At the lysosomes, proteases digest the contents 4.
- PTs release LMW products of digestion across the basolateral membrane into the peritubular circulation 5.
- Hardly any reabsorption of intact proteins.
- Small subset of proteins avoids the lysosomes and moves by transcytosis for release at the basolateral membrane.

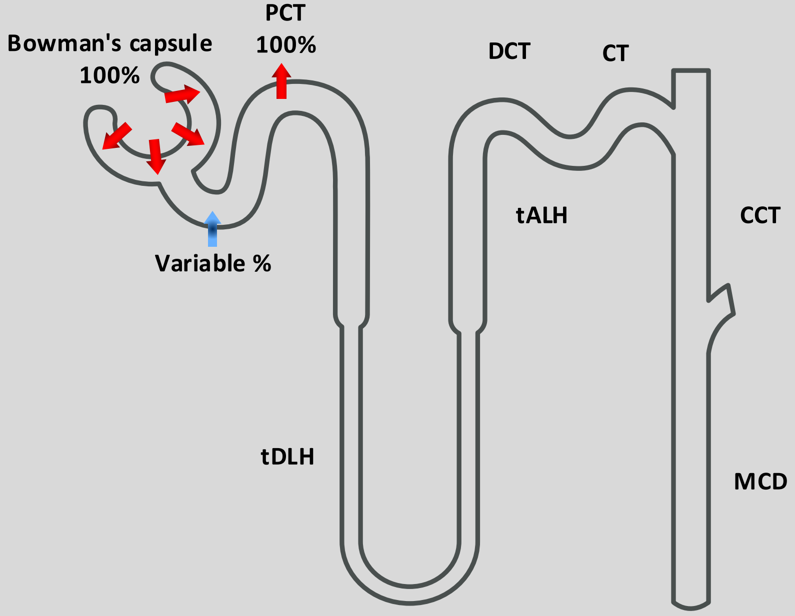




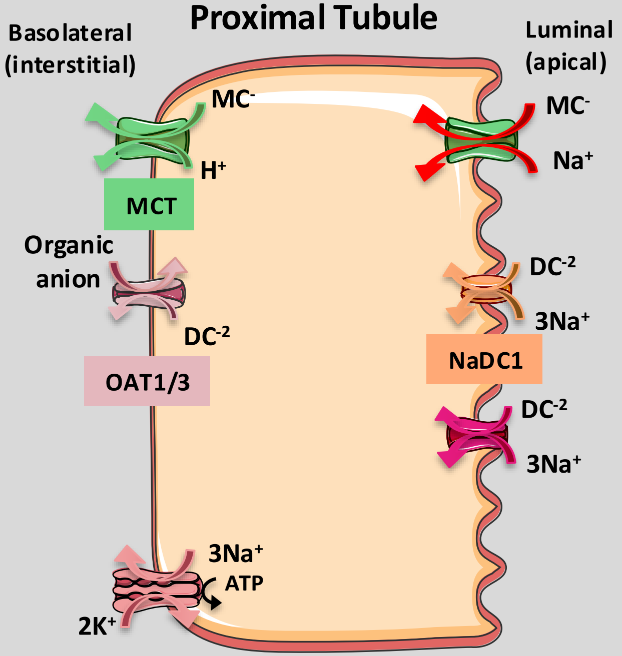
Organic Acid Secretory System (OASS)

- Mono- (R-COO$^-$), di- (R-2COO$^-$) & tricarboxylates (R-3COO$^-$): Products of Glc & fatty acid metabolism (anaerobic/citric acid cycle).
- Important for energy metabolism ∴ urine loss is wasteful.
- PT reabsorbs virtually all these substances.
- ↑ [Carboxylates]$_{Pl}$ → [Carboxylates]$_U$
- Urinary excretion occurs when the FL of ketone bodies (acetoacetate, β-OH butyrate) exceeds their T_m in PT.
- [Carboxylates]$_{Pl}$: 1-3 mM (lactate is the largest fraction).
- Na^+-dependent cotransporters carry carboxylates across the apical membranes
- All are members of the Solute Carrier (SLC) family of cotransporters.
- R-COO$^-$ exit across the basolateral membrane via MCT2.
- Efflux is coupled to H^+ → [Na^+]$_{IC}$ & ↑pH_{IC}.
- R-2COO$^-$ exit the cell across the basolateral membrane via the renal organic anion transporters (OAT1 & OAT3).
- OATs overlap in substrate specificity.
- Para-aminohippurate (PAH): Small (MW: 194 Da), H_2O-soluble organic anion.
- Freely filtered & secreted by the PT → More is removed than filtered.
- At low plasma concentrations, PAH entering the kidney is removed from the plasma and excreted.
- PAH clearance can be used experimentally as a measure of renal plasma flow (effective renal plasma flow).

Organic Acid Reabsorption in the Nephron



Proximal Tubule



Renal Regulation of Uric Acid

- Urate (Ur): Monovalent anion & the end product of purine catabolism

$$\left.\begin{array}{l}\text{Adenosine}\\ \text{Guanosine}\end{array}\right] \longrightarrow \text{Xanthine} \xrightarrow{\text{Xanthine Oxidase}} \text{URIC ACID} \longleftrightarrow \text{URATE ANION}$$

- $[Ur^-]_{Pl} \cong$ 3-7 mg/dL (0.2-0.4 mM)
- Filtered, reabsorbed & secreted (in PT).
- Reabsorption is the more important transport pathway.

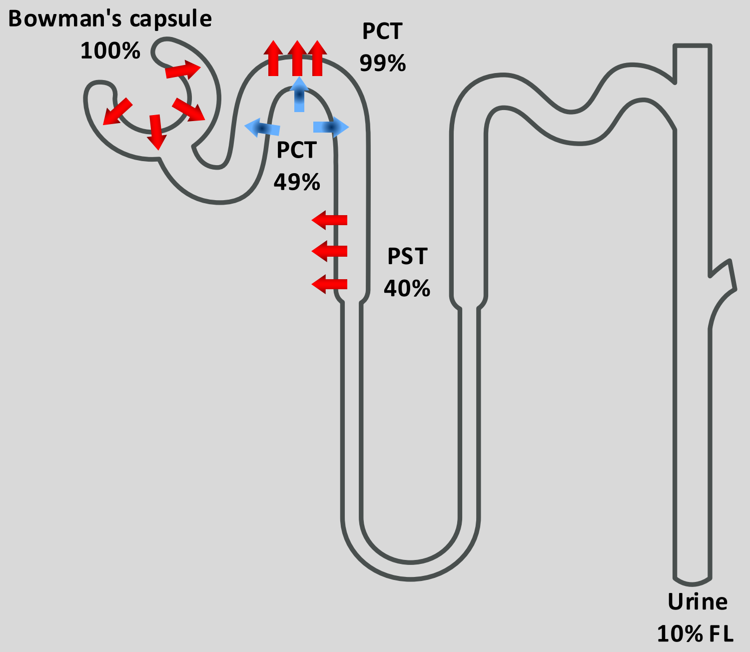
Reabsorption

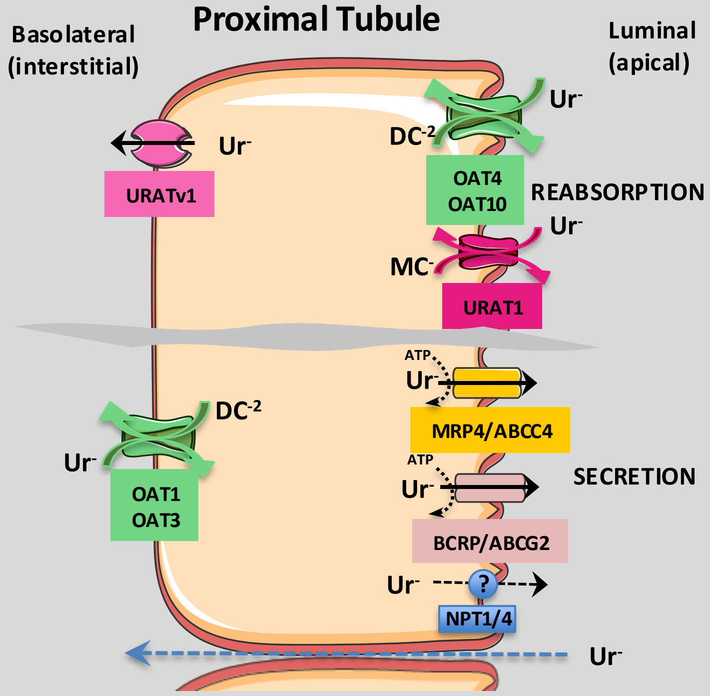
- PT reabsorbs Ur^- via paracellular (passive diffusion) & transcellular routes (active transport).
- Paracellular reabsorption:
  - Important during EC volume depletion
  - Results in compensatory enhancement of PT fluid reabsorption.
  - $\therefore \uparrow[Ur^-]_L \rightarrow \uparrow$ Paracellular Ur^- reabsorption $\rightarrow \downarrow Ur^-$ excretion $\rightarrow [Ur^-]_{Pl}$
- Transcellular reabsorption:
  - Apical membrane:
    - Mediated by URAT1, OAT4 & OAT10
  - Basolateral membrane:
    - Mediated by URATv1 (voltage-driven).

Secretion:

- Basolateral membrane: via OAT1 and OAT3
- Apical membrane: via NPT1/NPT4 & and two ABC transporters MRP4 & BCRP

Uric Acid Regulation in the Nephron





Organic Base Secretory System (OASS)

- PST (S3):
  - Responsible for secreting a wide range of endogenous and exogenous organic cations (organic bases).
- Important secreted <u>endogenous</u> organic cations:
  - Monoamine neurotransmitters (e.g., Dopa, Epi, NE, His)
  - Creatinine: the breakdown product of phosphocreatine.
- Important secreted <u>exogenous</u> organic cations:
  - Morphine, quinine, atropine…….
  - K^+-sparing diuretics: Amiloride, triamterene.
- Driving force for basolateral membrane movement of OC^+ is the inside-negative membrane potential.
- Via a polyspecific organic cation transporter OCT2.
- At the apical membrane, exchangers MATE1, MATE2-K & MDR1 move OC^+ from cell to the lumen.
- Dependent on the H^+ gradient across the apical membrane.

Organic Base Reabsorption in the Nephron